ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Cell: The Building Block of Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Robert Hooke	In thin slices of cork, 1665.
2	**(c)** prokaryotic cell	e.g. bacteria.
3	**(c)** rough ER	Protein synthesis.
4	**(c)** Schleiden and Schwann	Plants and animals are made of cells.
5	**(a)** bacterium	No nuclear membrane.
6	**(a)** lipid synthesis and detoxification	Fats and detox.
7	**(c)** new cells arise from pre-existing cells	All cells come from cells.
8	**(a)** mitochondrion	Releases energy.
9	**(b)** DNA and proteins	Hereditary material.
10	**(a)** unicellular	e.g. Amoeba.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Robert Hooke in 1665 (1); Box-like compartments in a cork slice (1).	1+1
12	Chloroplast: photosynthesis (1); Chromoplast: colour; leucoplast: storage (1).	1+1
13	Shrinking of the cytoplasm away from the cell wall (1); In a hypertonic solution (1).	1+1
14	All living organisms are made of cells (1); Cell is the basic unit; new cells arise from existing cells (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Raisin in water swells due to endosmosis (1); Raisin in strong sugar solution shrinks due to exosmosis (1); Water moves through the selectively permeable membrane (1).	1+1+1
16	Shape suits the function (1); RBC: disc-shaped, carries oxygen (1); Nerve cell: long and branched, carries messages (1).	1+1+1
17	Store water and nutrients (1); Maintain turgidity (1); Store waste products (1).	1+1+1
	OR: Hollow a potato and put sugar solution in it (1); Place in water (1); Water level in the cavity rises by osmosis (1).	1+1+1
18	Hypotonic: lower solute outside, cell swells (1); Hypertonic: higher solute outside, cell shrinks (1); Isotonic: equal, no change (1).	1+1+1
	OR: Water enters by osmosis (1); Cell wall resists the expansion (1); Cell becomes turgid but does not burst (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Cell membrane (1); Nucleus with chromosomes (1); Cytoplasm with ER and Golgi (1); Mitochondria, lysosomes (1); Ribosomes and vacuoles (1).	1+1+1+1+1
	OR: Mitosis for growth and repair (1); Meiosis for gamete formation (1); Chromosome numbers (1); Daughter cells (1); Significance (1).	1+1+1+1+1
20	Cell wall (1); Chloroplasts and other plastids (1); Large central vacuole (1); Common organelles (1); Functions (1).	1+1+1+1+1
	OR: Mitochondria release energy from food (1); They are called the powerhouse of the cell (1); Chloroplasts contain chlorophyll for photosynthesis (1); Chromoplasts give colour to flowers and fruits (1); Leucoplasts store starch, oils and proteins (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that bacteria have a nucleus, or that prokaryotes have no genetic material; the DNA is present but is not enclosed in a nuclear membrane.
- Mixing up hypotonic and hypertonic: in a *hypotonic* outside solution the cell swells, in a *hypertonic* one it shrinks.
- Calling osmosis the movement of any solute; osmosis is the movement of *water* through a selectively permeable membrane.

- Confusing the functions of rough ER (protein synthesis, studded with ribosomes) and smooth ER (lipids, detoxification), or saying that mitochondria 'make' energy; they release energy from food as ATP.
- Saying that the cell wall and cell membrane are the same; the membrane is selectively permeable and present in all cells, the wall is rigid and fully permeable.
- Forgetting that plant cells do not burst in water because of the cell wall, whereas an animal cell like RBC can swell and burst.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Cell: The Building Block of Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Golgi apparatus	Golgi.
2	**(a)** DNA	Heredity units.
3	**(a)** Paramecium	One cell.
4	**(d)** lysosomes	Digestive enzymes.
5	**(b)** swells	Water enters.
6	**(c)** allows only certain substances to pass	Controls entry and exit.
7	**(b)** ribosomes	Ribosomes.
8	**(b)** shrinks (plasmolysis)	Water leaves.
9	**(a)** diffusion	Net movement down a gradient.
10	**(d)** chloroplasts	Photosynthesis.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Prokaryotes have no nuclear membrane; eukaryotes do (1); Prokaryotes lack membrane-bound organelles (1).	1+1
12	Digest worn-out organelles and foreign matter (1); Called suicide bags (1).	1+1
13	Diffusion: movement of any particles from higher to lower concentration (1); Osmosis: movement of water through a selectively permeable membrane (1).	1+1
14	Release energy from food in respiration (1); Energy stored as ATP (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Thin flexible layer of lipids and proteins (1); Selectively permeable (1); Controls the entry and exit of substances (1).	1+1+1
16	Mitosis: growth, repair, 2 identical cells (1); Meiosis: gametes, 4 cells with half chromosomes (1); Chromosome number same vs halved (1).	1+1+1
17	Nuclear membrane: separates nucleus (1); Nucleolus: makes ribosomes (1); Chromatin: contains DNA (1).	1+1+1
	OR: Cell wall present only in plant cells (1); Plastids only in plants (1); Vacuoles large in plants, small in animals (1).	1+1+1
18	Bodies are made of cells (1); Each cell carries out life processes (1); Cells arise from pre-existing cells (1).	1+1+1
	OR: (a) Mitochondria (1); (b) Chloroplast; (c) Ribosomes (1); (d) Lysosomes; (e) Nucleus (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Diffusion of ink or smell (1); Osmosis: raisins or potato (1); Water uptake by roots (1); Exchange of gases (1); Definitions (1).	1+1+1+1+1
	OR: RBC (1); Nerve cell (1); Muscle cell (1); Root hair cell (1); Guard cell (1).	1+1+1+1+1
20	Nucleus (1); Size (1); Organelles (1); DNA (1); Examples (1).	1+1+1+1+1
	OR: Nuclear membrane, nucleolus (1); Chromosomes in the nucleus (1); DNA carries information (1); Genes are segments (1); Heredity (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that bacteria have a nucleus, or that prokaryotes have no genetic material; the DNA is present but is not enclosed in a nuclear membrane.
- Mixing up hypotonic and hypertonic: in a *hypotonic* outside solution the cell swells, in a *hypertonic* one it shrinks.
- Calling osmosis the movement of any solute; osmosis is the movement of *water* through a selectively permeable membrane.
- Confusing the functions of rough ER (protein synthesis, studded with ribosomes) and smooth ER (lipids, detoxification), or saying that mitochondria 'make' energy; they release energy from food as ATP.
- Saying that the cell wall and cell membrane are the same; the membrane is selectively permeable and present in all cells, the wall is rigid and fully permeable.

- Forgetting that plant cells do not burst in water because of the cell wall, whereas an animal cell like RBC can swell and burst.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Cell: The Building Block of Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** swells up	Osmosis into the raisin.
2	**(a)** osmosis	Water diffusion.
3	**(b)** chromoplasts	Coloured pigments.
4	**(d)** shows no net movement of water	Equal concentrations.
5	**(a)** nucleus	Contains DNA.
6	**(c)** leucoplasts	Store food.
7	**(b)** mitosis	Growth and repair.
8	**(c)** cellulose	Rigid layer.
9	**(d)** mature plant cells	Store water and sap.
10	**(a)** gametes with half the chromosome number	Reduction division.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Plant cells have a cell wall and plastids (1); Plant cells have a large vacuole (1).	1+1
12	Contains DNA/chromosomes (1); Controls all the cell activities and heredity (1).	1+1
13	Protects the cell contents (1); Controls the entry and exit of materials (1).	1+1
14	Surface area to volume ratio falls as cell grows (1); Materials cannot be exchanged efficiently (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	ER: transport and synthesis of proteins and lipids (1); Golgi: modifies, packages and dispatches materials (1); They work together in vesicles (1).	1+1+1
16	Nucleus: controls the cell (1); Mitochondrion: releases energy (1); Ribosome: makes proteins (1).	1+1+1
17	Nucleus (1); Organelles (1); Size (1).	1+1+1
	OR: Mitochondria: energy release (1); Ribosomes: protein synthesis (1); Lysosomes: digestion (1).	1+1+1
18	Growing cells outside the body in a nutrient medium (1); Production of vaccines (1); Tissue culture in plants (1).	1+1+1
	OR: Plants need rigid support (1); Animals have skeletons and muscles (1); Animal cells need flexibility (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Schleiden, Schwann (1); Virchow (1); Basic unit (1); Viruses are acellular (1); Modern additions (1).	1+1+1+1+1
	OR: Growing cells (1); In nutrient medium (1); Vaccine production (1); Micropropagation (1); Stem cells (1).	1+1+1+1+1
20	Hypotonic: plant cell turgid, RBC swells and may burst (1); Hypertonic: plasmolysis, RBC shrinks (1); Isotonic: no change (1); Role of the cell wall (1); Examples (1).	1+1+1+1+1
	OR: ER makes (1); Golgi packages (1); Vesicles carry (1); Lysosomes digest (1); Coordination (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that bacteria have a nucleus, or that prokaryotes have no genetic material; the DNA is present but is not enclosed in a nuclear membrane.
- Mixing up hypotonic and hypertonic: in a *hypotonic* outside solution the cell swells, in a *hypertonic* one it shrinks.
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- Confusing the functions of rough ER (protein synthesis, studded with ribosomes) and smooth ER (lipids, detoxification), or saying that mitochondria 'make' energy; they release energy from food as ATP.
- Saying that the cell wall and cell membrane are the same; the membrane is selectively permeable and present in all cells, the wall is rigid and fully permeable.
- Forgetting that plant cells do not burst in water because of the cell wall, whereas an animal cell like RBC can swell and burst.

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